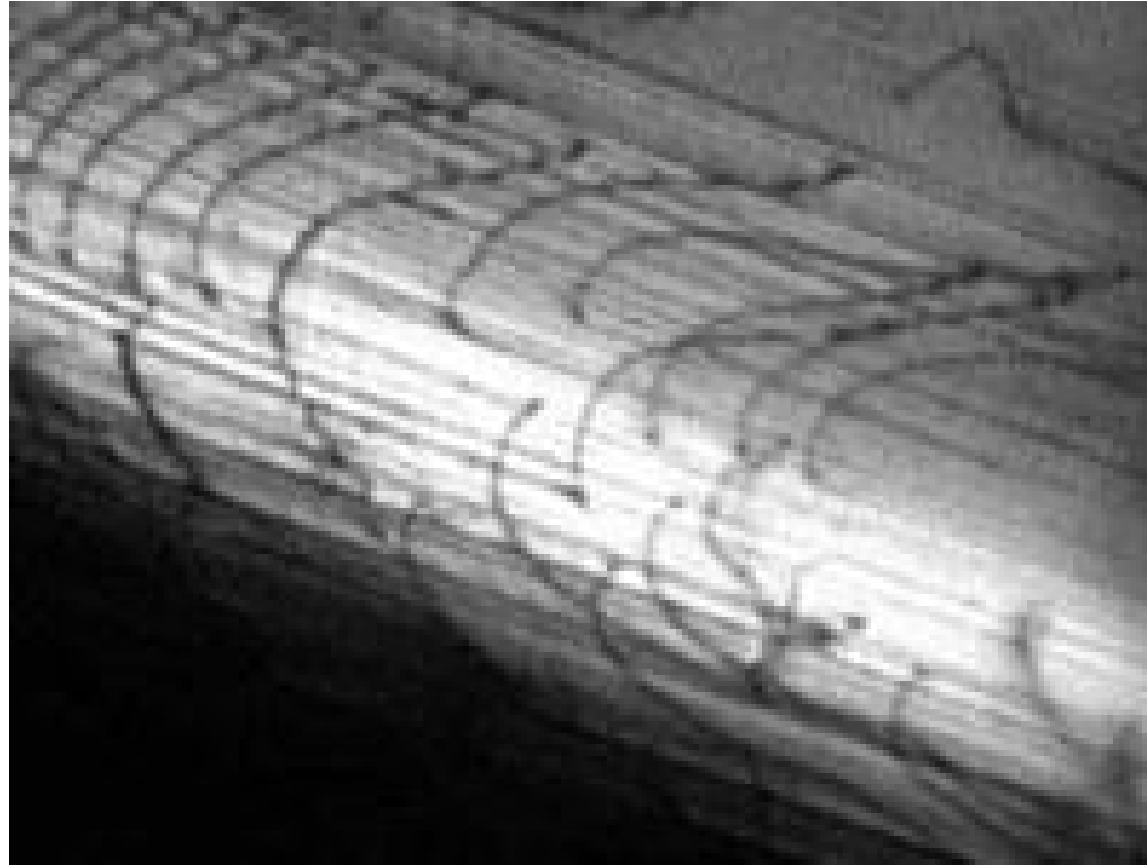


APS110 Tutorial #6

Daniel Brlas

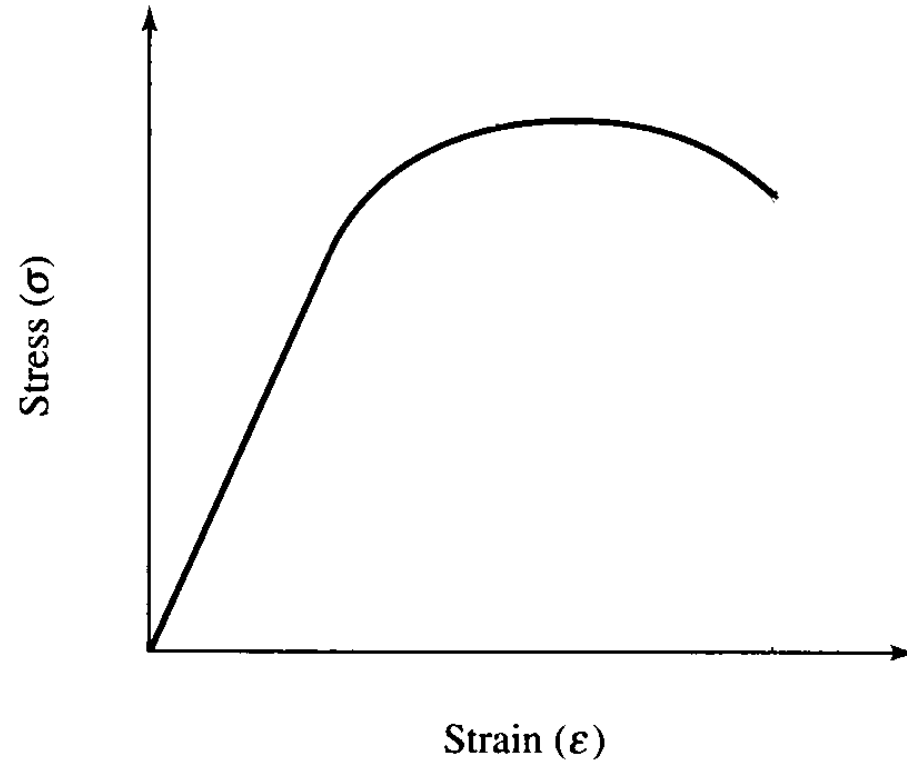
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Review – Stress-Strain Curves

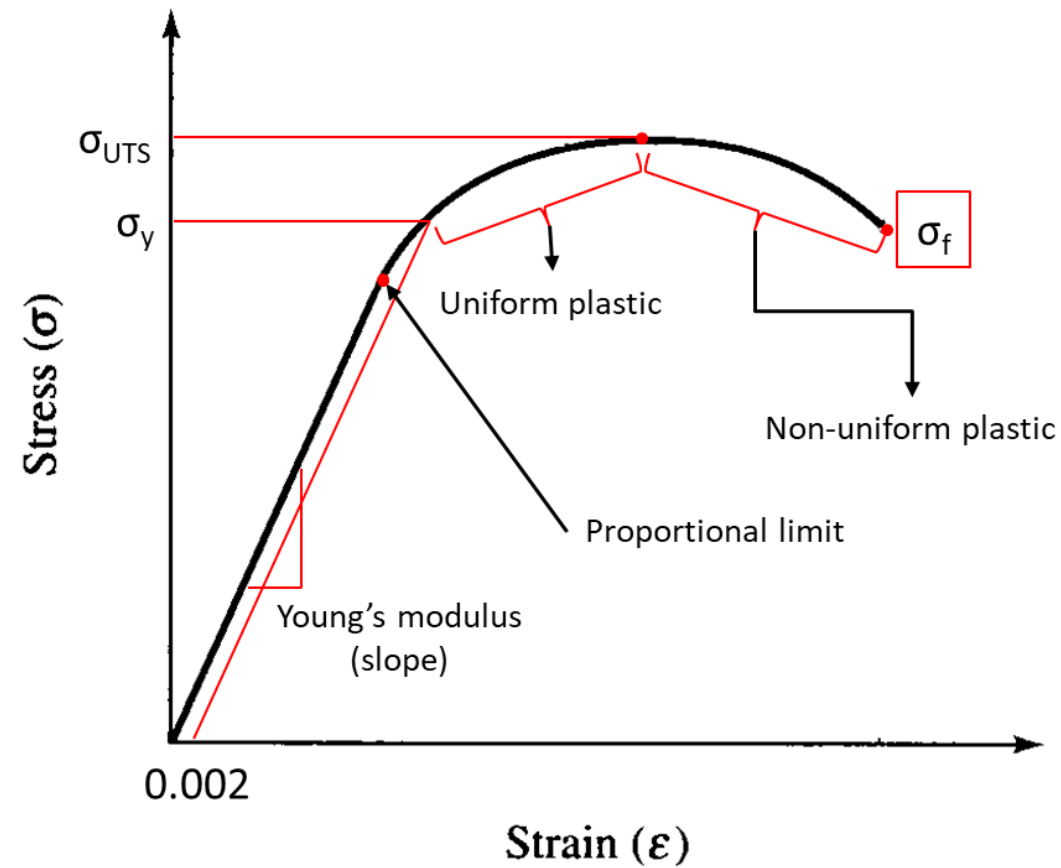


Practice Question 1

1. On the generalized stress and strain curve for a typical metal as presented below, clearly identify each of the following characteristics:
- a) 0.2% offset yield strength
 - b) Young's modulus
 - c) Proportional limit
 - d) Ultimate tensile strength
 - e) Region of non-uniform plastic deformation
 - f) Region of uniform plastic deformation
 - g) Fracture strength

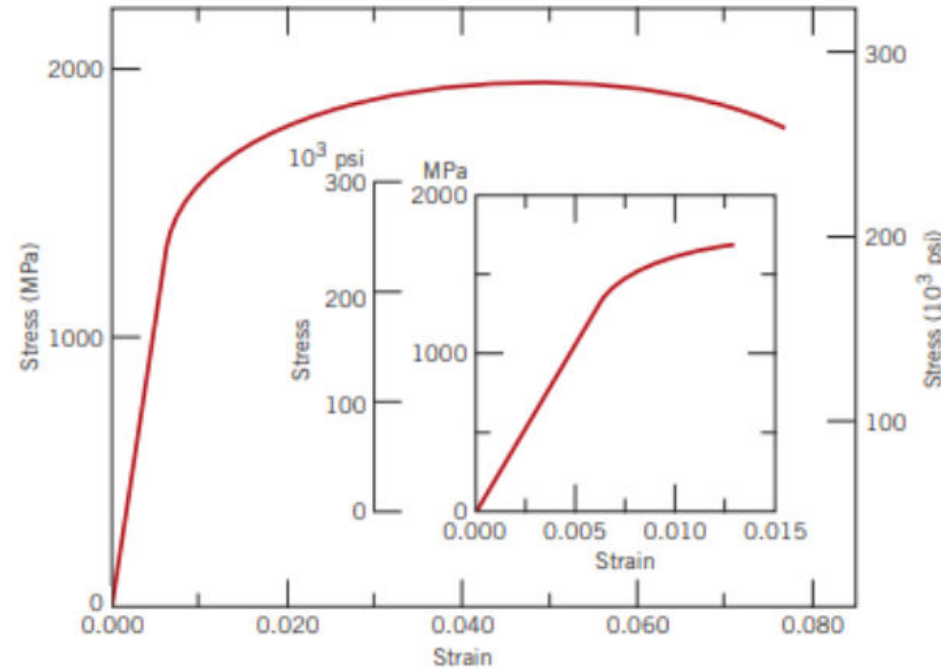


Practice Question 1

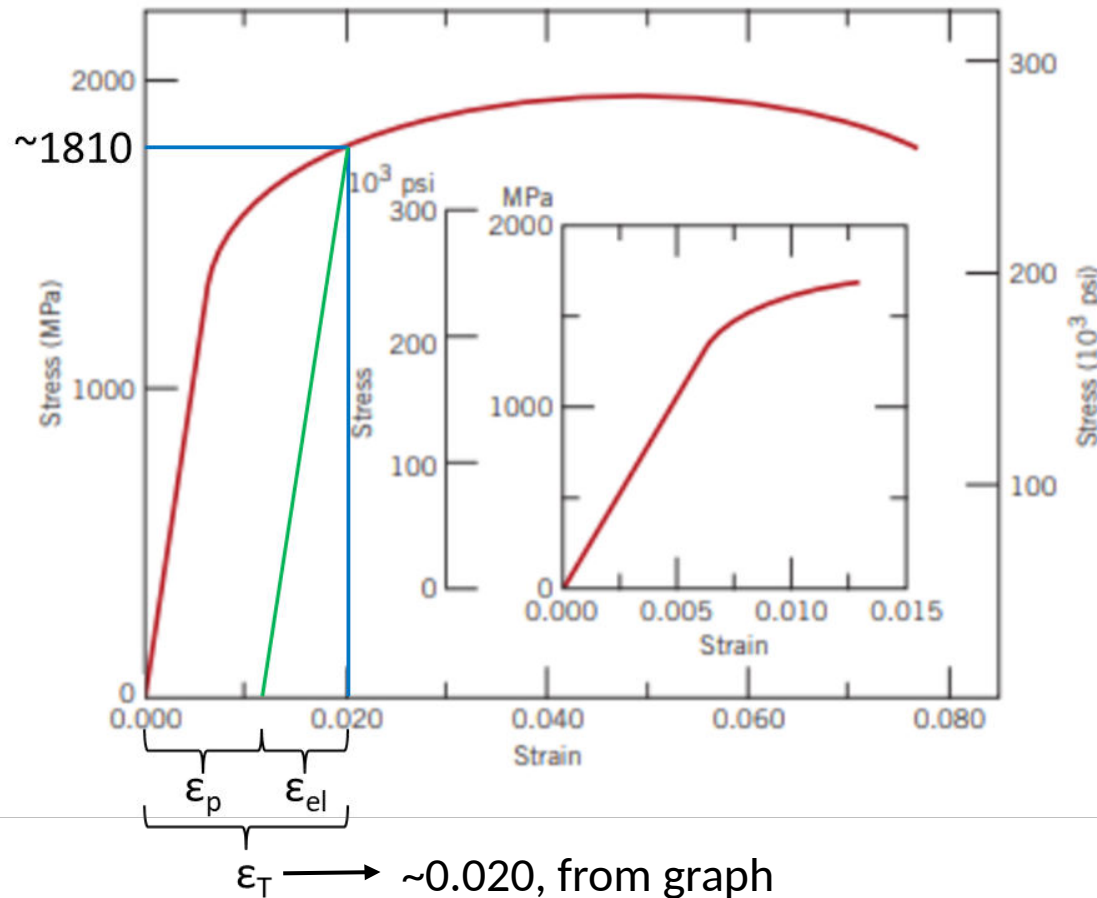


Practice Question 2

2. Below is the same stress-strain curve given in the problem set. Assume now that the behaviour below is for a sample with a cross-sectional width and depth of 19 mm by 3.2 mm and 610 mm long. Determine the following when a force of 110kN is applied onto a steel sample with a Young's modulus of 207 GPa.
- a) Elastic strain and total strain
 - b) Plastic strain
 - c) Sample length after unloading



Practice Question 2



$$\sigma = \frac{F}{A_0} = \frac{110\,000}{(19e^{-3})(3.2e^{-3})} = 1810 \text{ MPa}$$

$$\varepsilon_{el} = \frac{\sigma}{E} = \frac{1810e6}{207e9} = 0.0087$$

$$\varepsilon_p = \varepsilon_T - \varepsilon_{el} = 0.020 - 0.0087 = 0.0113$$

$$\varepsilon = \frac{l_i - l_0}{l_0} \rightarrow l_i = l_0(1 + \varepsilon_p)$$

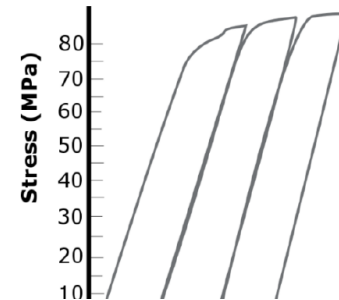
Note: We use ε_p at the end as it corresponds to our remaining deformation after unloading the sample

$$l_i = 610(1 + 0.0113) = 616.7 \text{ mm}$$

Note: We use ε_p at the end as it corresponds to our remaining deformation after unloading the sample

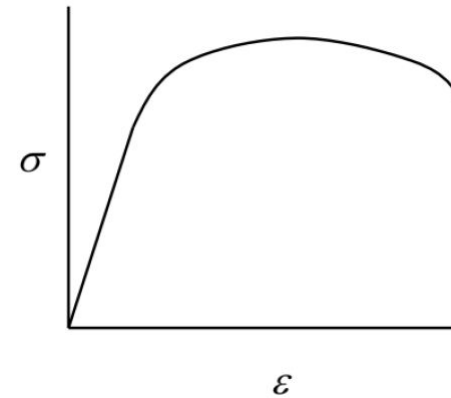
Practice Quiz

1. The Young's modulus is related to which of the following:
 - a) The slope of the interatomic force separation curve taken at the equilibrium interatomic spacing
 - b) The slope of the interatomic energy separation curve taken at the equilibrium interatomic spacing Region between the ultimate tensile strength and fracture
 - c) The slope of the interatomic force separation curve taken at $r=0$
 - d) Both a) and c)
2. Which of the following most closely describes the beginning of non-uniform plastic deformation in a metallic sample?
 - a) The proportional limit
 - b) The 0.2% offset yield strength
 - c) The ultimate tensile strength
 - d) None of the above
3. In the tensile testing results below, what is the final plastic strain experienced by the sample?
 - a) 0.045
 - b) 0.032
 - c) 90 MPa
 - d) 0.013

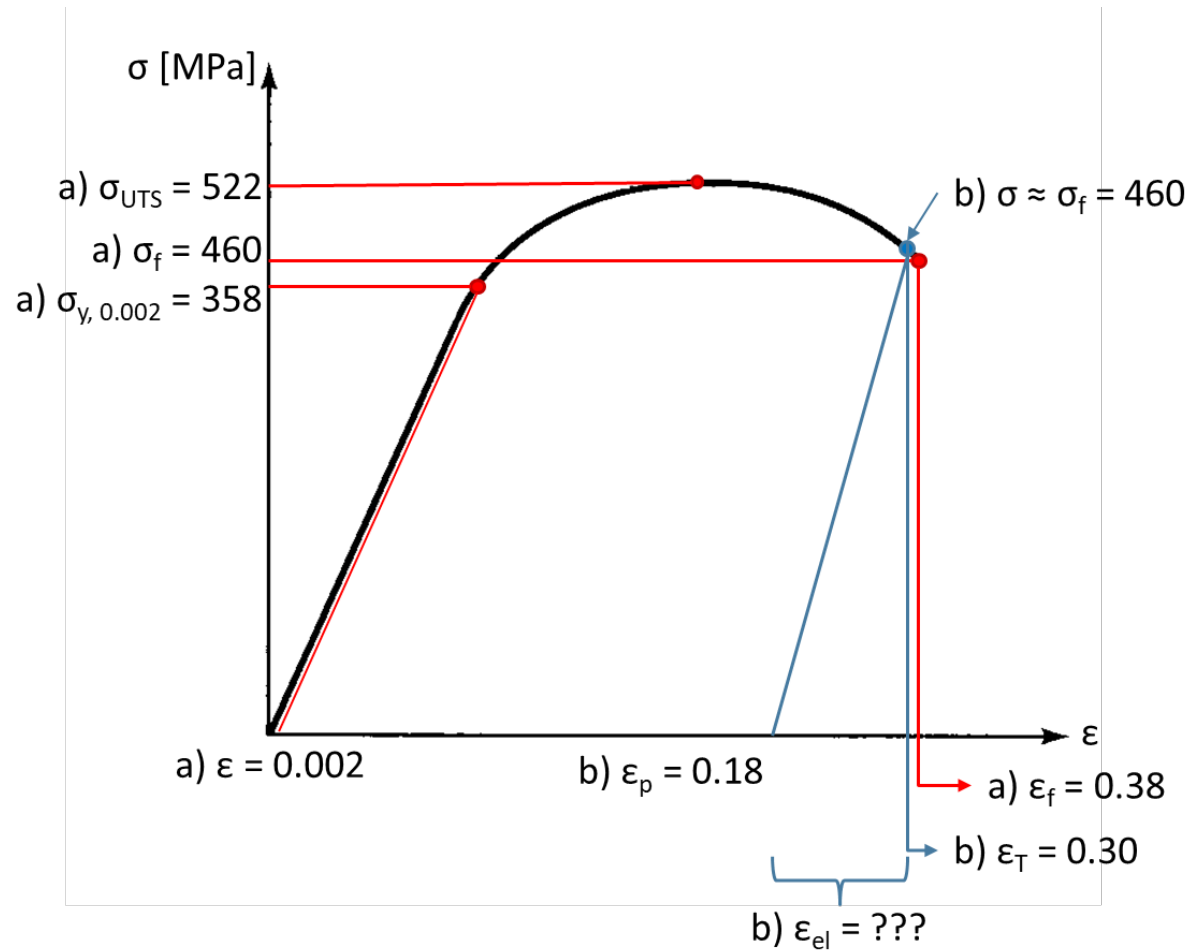


Practice Quiz

4. A metallic sample is loaded in tension and exhibits the stress-strain behaviour shown below. Which of the following is the best description of why the sample failed at the location where the neck first formed?
- a) Because the neck always forms in the middle of the reduced section
 - b) Because the metal got weaker when it necked
 - c) Because the metal did not strengthen fast enough to account for the local decrease in cross-sectional area
 - d) Because the metal got weaker and the cross-sectional area decreased
 - e) None of the above



5. A hypothetical metal has a 0.2% offset yield strength of 358 MPa, an ultimate tensile strength of 522 MPa, and a fracture strength of 460 MPa at a fracture strain of 0.38
- a) Sketch the stress-strain curve based on the provided information
 - b) A sample of this metal, originally 1m in length with a cross section of 2 mm x 2 mm is loaded along its long axis. Just before fracture, while the load is still applied, the length is 1.3m and



b)

As the question states, the sample is loaded to just before fracture, so you can estimate the applied stress to be around the fracture stress of 460 MPa.

Find plastic and total strain using: $\epsilon = \frac{\Delta l}{l_0}$, then

$$\epsilon_{el} = \epsilon_T - \epsilon_p = 0.30 - 0.18 = 0.12$$

$$E = \frac{\sigma}{\epsilon_{el}} = \frac{460 \text{e}6}{0.12} = 3.8 \text{ GPa}$$